温州肯恩大学
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MGS 3001 WHS01
Python Programming for Business

Hypothesis & Measurement

Research Workshop II

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Learning Objectives

By the end of this workshop, you should be able to:

- Refine your research question into a testable research question
- Formulate **hypotheses** about relationships between variables
- Distinguish constructs from variables
- Design **measurement** to operationalize constructs
- Align research ambitions with practical **data availability**

Research Pipeline

1. Research Question

What do you want to know? The starting point of inquiry.

E.g. Does remote work affect productivity?

2. Hypothesis

What do you predict? Your testable statement.

E.g. Remote work increases productivity

3. Measurement

How will you capture concepts? Operationalization.

E.g. Tasks completed per hour

4. Data Collection

Gathering empirical evidence systematically.

E.g. Survey 500 remote workers

5. Data Analysis

Testing hypotheses and drawing conclusions.

E.g. Regression analysis

From curiosity to knowledge: the five-stage research process

Do You Always Need Hypotheses?

✓ With Hypotheses

Confirmatory Research
Testing existing theories or predictions

Causal Analysis
Establishing cause-effect relationships

Theory Testing
Validating or refuting theoretical claims

→ Structured, focused approach

Without Hypotheses

Exploratory Research
Investigating new phenomena without preconceptions

Descriptive Studies
Documenting patterns, trends, or characteristics

Grounded Theory
Building theory from data inductively

→ Open-ended discovery

Hypotheses are **NOT mandatory** but highly recommended for most quantitative research.

When to Use Hypotheses

Clear Relationship

You have strong theoretical or empirical reasons to expect a specific relationship between variables.

"Based on existing theory, we expect X to influence Y"

Mechanism Exists

You can articulate **how** and **why** the relationship occurs—the causal pathway is clear.

"X affects Y through mechanism Z"

Testing Theory

Your goal is to validate, refine, or challenge existing theoretical predictions.

"This study tests Theory T's prediction about..."

When Not to Use Hypotheses

Exploratory

You're entering uncharted territory where little is known. Premature hypotheses may blind you to unexpected patterns.

E.g. *Studying social media's impact on teenage mental health in 2005 (before research existed)*

New Phenomena

Emerging trends, technologies, or social changes where theory hasn't caught up with practice.

E.g. *Early research on cryptocurrency adoption, pandemic remote work effects*

Pattern Discovery

Your goal is to identify patterns, generate insights, or map a domain before testing specific claims.

E.g. *Clustering customer segments, mapping network structures*

The Golden Rule

- When in doubt, ask: *"Do I know enough to make a specific prediction?"*
- If the answer is no, start exploratory.
- You can always develop hypotheses after initial findings.

Why Hypotheses Matter

Theory
Literature & Logic

Hypothesis
Testable Prediction

Data
Empirical Evidence

Testable Predictions

Hypotheses transform abstract theories into concrete, falsifiable statements.

They specify what you expect to find, making your research focused and rigorous.

Link Theory → Data

Hypotheses serve as the crucial bridge between theoretical concepts and empirical observation.

They operationalize abstract ideas into measurable relationships.

Guide Analysis

Clear hypotheses determine your analytical strategy – what statistical tests to run, what control variables to include, and how to interpret findings.

Build Logic First

STEP 1 **Phenomenon**

Observe a pattern, trend, or puzzle in the world that sparks your curiosity. What caught your attention?

↓

STEP 2 **Mechanism**

Identify the causal pathway. **How** and **why** does this phenomenon occur? What drives the relationship?

↓

STEP 3 **Literature**

What do existing theories and studies say? Build on prior knowledge to ground your prediction in established scholarship.

↓

STEP 4 **Prediction**

Synthesize into a specific, testable statement. Given the phenomenon, mechanism, and literature, what do you predict will happen?

Example: AI and Workplace Productivity

The Logic Chain

1. **Phenomenon:** AI tools are increasingly adopted in workplaces
2. **Mechanism:** AI automates routine tasks → reduces cognitive effort → frees up time
3. **Literature:** Technology adoption theory suggests efficiency gains
4. **Prediction:** AI use should improve productivity metrics

The Hypothesis

Employees who use AI tools complete tasks faster than those who don't, controlling for task complexity.

- **Testable:** Can measure task completion time
- **Relational:** Compares AI users vs. non-users
- **Specific:** Predicts improvement

What Is a Hypothesis?

AFFECTS

Independent variable (IV) → Dependent variable (DV)

Testable

A hypothesis must be empirically verifiable.

You must be able to collect data that could potentially support or refute it.

✓ **Good:** AI reduces task time by 15%

✗ **Bad:** *AI makes work better (vague)*

Relational

It proposes a connection between two or more variables – how they co-vary or influence each other.

✓ **Good:** Training increases performance

✗ **Bad:** *Training is important (no relation)*

Theory-Driven

Grounded in existing knowledge, literature, or logical reasoning – not just a random guess or personal hunch.

✓ **Good:** Based on human capital theory...

✗ **Bad:** *I think this might happen...*

How to Establish a Strong Hypothesis

1 Identify Constructs

What are the abstract concepts in your research question?

E.g. *Productivity, job satisfaction, innovation*

2 Specify Relationship

How do these constructs relate? Direction and magnitude?

E.g. *Increases ~~by 10-15%~~*

3 Define Context

Where, when, and for whom does this apply?

E.g. *Knowledge workers in tech firms*

4 Ensure Measurable

Can you operationalize these constructs with available data?

E.g. *Tasks completed per hour*

Example: Remote Work & Productivity

1. **Constructs:** Remote work (IV) → Productivity (DV)
2. **Relationship:** Increases ~~by 10-13% (Bloom et al., 2015)~~
3. **Context:** Call center workers at Ctrip, China
4. **Measurable:** Calls handled per shift (company records)

Final Hypothesis:

Remote work increases call center productivity ~~by 10-13%, measured as calls handled per shift,~~ compared to office-based work.

Common Hypothesis Mistakes

1. Vague Language

Using words like "affects," "impacts," or "influences" without specifying direction or magnitude. These are relationship weasel words.

❌ **Weak:** "Training affects performance"
What kind of effect? How much?

✓ **Strong:** "Training increases sales ~~by 12%~~"
Clear direction and magnitude

2. No Direction

Failing to specify whether the relationship is positive or negative. A hypothesis must make a directional prediction (unless truly exploratory).

❌ **Weak:** "Competition and innovation are related"
Related how? Positive or negative?

✓ **Strong:** "Competition increases innovation ~~by 8%~~"
Clear positive relationship

3. Not Measurable

Proposing constructs that cannot be operationalized with available data or methods. If you can't measure it, you can't test it.

❌ **Weak:** "Leadership improves organizational culture"
"Culture" is abstract—how to measure?

✓ **Strong:** "Leadership training reduces turnover ~~by 15%~~"
Turnover is measurable from HR records

From Hypothesis to Measurement

Abstract Prediction

“AI improves productivity”

Hypothesis

Transform

→

Operationalization

Measurement

Concrete Variables

“Tasks per hour”

Constructs

Abstract theoretical concepts that exist in our minds but cannot be directly observed.

E.g. Intelligence, motivation, organizational culture, job satisfaction

Operationalization

The process of defining constructs in terms of observable, measurable operations.

Process: Concept → Indicators → Variables → Data

Variables

Concrete, measurable representations of constructs that can take different values.

E.g. IQ score, Likert scale rating, turnover rate, revenue

Measurement is the bridge that connects your theoretical hypothesis to empirical reality. Without it, your hypothesis remains untestable.

Constructs vs. Variables

Construct

the abstract concept

- Exists in theory
- Cannot be directly observed
- Defined conceptually
- Multi-dimensional

Variable

the measurable proxy

- Exists in data
- Directly observable
- Defined operationally
- Takes numeric values

Concrete Example: Productivity

CONSTRUCT

Productivity

Abstract concept: efficiency in producing output

→

VARIABLE

Time per Task

Measurable: minutes to complete one unit

From Construct to Data

Level 1
Productivity
Abstract construct

→

Level 2
Efficiency
Sub-dimension

→

Level 3
Time/Task
Indicator

→

Level 4
Dataset
[23, 45, 31...]

The Operationalization Process

1. **Define the Construct**
   Conceptualize what you mean
2. **Identify Dimensions**
   Break into components
3. **Select Indicators**
   Choose observable proxies
4. **Collect Data**
   Gather numeric values

Example: Job Satisfaction

Construct: Job Satisfaction (overall attitude)

Dimensions: Pay, work, colleagues, growth

Indicators: Survey questions (1-7 scale)

Data: [5, 6, 4, 7, 3...] from 500 employees

Reflective vs. Formative Measures

Reflective

- Indicators are *effects* of the construct.
- They reflect the underlying latent variable.

Construct → Ind. 1 Ind. 2 Ind. 3

✓ Indicators are interchangeable
✓ Removing one doesn't change meaning
✓ Internal consistency expected *(Cronbach's alpha)*

Example: Satisfaction → [happy with pay, work, colleagues]

Formative

- Indicators *form or define* the construct.
- They form the composite variable.

Ind. 1 Ind. 2 Ind. 3 → **Construct**

✓ Indicators are NOT interchangeable
✓ Removing one changes the meaning
✓ Internal consistency NOT expected

Example: [sadness, fatigue, loss of interest] → Depression

Most psychological traits are reflective;
most socioeconomic indices are formative.

Why Measurement Is Difficult

Abstract Nature

- Constructs like “motivation” or “culture” have no physical existence.
- We must infer them from behaviors and responses.

Challenge: How do we capture something we can't directly see?

Limited Data

- Available data may not perfectly capture your construct.
- You often must use proxies that approximate what you really want.

Challenge: What if ideal data doesn't exist?

Requires Judgment

- There's rarely one “right” way to measure a construct.
- You must make reasoned choices and defend them.

Challenge: How do you justify your choices?

The Reality

- All measurement involves some degree of approximation.
- Your goal is not perfection but transparent, defensible choices that minimize error and bias.

Measurement Approaches

1. Literature

Use measures that have been validated and published in peer-reviewed research.

Best For:

Established constructs with extensive research

- ✓ High credibility
- ✓ Proven reliability
- ✓ Easy to justify

2. Adapt

Modify existing measures to fit your specific context, population, or research needs.

Best For:

Similar but not identical contexts

- ✓ Balances validity & fit
- ✓ Builds on prior work
- ⚠ Requires justification

3. New

Develop original measures when no suitable existing measures are available.

Best For:

Novel constructs or phenomena

- ✓ Custom fit
- ⚠ Requires validation
- ⚠ Higher scrutiny

Recommendation Hierarchy

- For undergraduate research: **1. Literature > 2. Adapt > 3. New**.
- Always start by searching for validated measures.
- Only create new measures if absolutely necessary, and be prepared for additional validation work.

Defending Your Measurement

1 Why Is It Valid?

Does your measure actually capture the construct you claim? Establish face validity, content validity, and construct validity.

Face Validity	***Content Validity***	***Construct Validity***
Looks right on surface	*Covers all dimensions*	*Matches theory*

2 Why Is It Reasonable?

Given your constraints (time, budget, data access), is this the most practical approach? Acknowledge limitations transparently.

Example: *"We use self-reported productivity because objective company data was unavailable. While subject to bias, this approach is standard in the literature and allows us to..."*

3 Why Is It Better Than Alternatives?

What other measures did you consider? Why did you reject them? Show you've thought critically about options.

Example: *"We chose output per hour over self-rated productivity because (1) it's objective, (2) it aligns with economic theory, (3) prior studies show it's less prone to bias."*

Data Feasibility

Access Constraints

Company data, personal records, or proprietary information may be inaccessible due to privacy, legal, or competitive concerns.

Cost Constraints

Surveys, experiments, or data purchases can be expensive. Your budget may limit sample size or data quality.

Time Constraints

Longitudinal data collection or complex experiments may exceed your project timeline.

The Feasibility Principle

- Your research question and hypotheses must fit within your data constraints.
- **Never design a study you cannot execute.**
- Start with available data, then craft questions that can be answered with that data.

Robustness Checks

stability of the conclusion

Test whether your **main conclusion still holds under alternative specifications or methods**

- Alternative model specifications
- Different samples
- Different measurement choices
- Alternative estimation methods

Measure A → **AND** **Measure B** → **AND** **Measure C**

Self-reported productivity — Output per hour — Manager rating

Same Result Across All Measures = Robust Finding

Why Multiple Measures?

- **Triangulation**: Converging evidence from different sources increases confidence
- **Bias Reduction**: Different measures have different biases – they may cancel out
- **Comprehensive Capture**: Complex constructs have multiple dimensions

Checking Consistency

1. Run analysis with Measure A – record results
2. Run analysis with Measure B – record results
3. Compare: Do conclusions remain the same?
4. Report consistency or explore why results differ

Sensitivity Analysis

magnitude and direction of change

Examines **how results change when you vary assumptions, parameters, or inputs**

- Narrow (parameter-focused)
- One type of robustness check

The Core Idea

Model
Your analysis

→

Variations
Small changes

→

Same Result
Robust finding

What to Vary

Variable Definitions
Change how you code or categorize variables

Sample Restrictions
Include/exclude certain subgroups

Model Specifications
Add/remove control variables

Example: Remote Work Study

Base: Remote work increases productivity by 13%

Sensitivity 1: Exclude part-time workers → 12%

Sensitivity 2: Different productivity measure → 14%

Sensitivity 3: Add experience control → 11%

✓ Finding is robust: 11-14% range consistently positive

In-Class Exercise

Research Question

Write your RQ here...

✓ *Is it focused?*
✓ *Is it testable?*

Hypothesis

Write your H here...

✓ *Is it directional?*
✓ *Is it specific?*

Measurement

Define your measures...

✓ *How will you operationalize?*
✓ *What's your source?*

Data

Describe your data...

✓ *Is it accessible?*
✓ *Is it sufficient?*

Exercise Instructions

1. Take out your research proposal draft
2. Fill in each box with your current thinking
3. Identify gaps or weak connections
4. Revise to strengthen alignment
5. Be prepared to share with a partner

- ✓ Clear logical flow from RQ → H → Measurement → Data
- ✓ Hypothesis is specific, directional, and testable
- ✓ Measures are operationalized with clear justification
- ✓ Data is feasible and accessible

Good Research

=

Clear + **Measurable** + **Testable**

Clear

Precise questions and predictions

Measurable

Operationalized constructs

Testable

Feasible data collection

Master these principles, and you'll produce research that is rigorous, credible, and impactful.

Questions? Let's discuss your proposals!